CLASSIFICATION OF YONEDA EXTENSIONS BY DERIVED EXT WITHOUT ENOUGH PROJECTIVES OR INJECTIVES

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Editorial note. No assumption of enough injectives or projectives is made. The original common-extension construction, including the concluding elementary commutative-diagram verification, is retained.

Editorial scope. One original target with its local construction proved in full. Classical category primitives are explicitly allowed.

Allowed standard prerequisites (editorial).

- Derived-category calculus of fractions. Morphisms in D(A) are roofs in the homotopy category with quasi-isomorphism denominators; equality admits a common denominator after homotopy adjustments. Standard good truncations and pushouts in an abelian category may be used.

Source reference. Original section 04VB: focused-seeds/ACL/sources/stacks-snapshot-20261011/categories.tex, source lines 3893–3899. Standard localization roofs, not the target extension classification. Stable Stacks tag displayed.

Definition 0.1. Let $\mathcal{A}$ be an abelian category. Let $A, B \in \mathrm{Ob}(\mathcal{A})$. For $i \geq 1$ a degree i *Yoneda extension* of B by A is an exact sequence

$$E : 0 \to A \to Z_{i-1} \to Z_{i-2} \to \ldots \to Z_0 \to B \to 0$$

in $\mathcal{A}$. We say two Yoneda extensions E and E' of the same degree are *equivalent* if there exists a commutative diagram

$$\begin{array}{ccccccccccc}
0 & \longrightarrow & A & \longrightarrow & Z_{i-1} & \longrightarrow & \ldots & \longrightarrow & Z_0 & \longrightarrow & B & \longrightarrow & 0 \\
 & & \uparrow{\scriptstyle \mathrm{id}} & & \uparrow & & & & \uparrow & & \uparrow{\scriptstyle \mathrm{id}} & & \\
0 & \longrightarrow & A & \longrightarrow & Z''_{i-1} & \longrightarrow & \ldots & \longrightarrow & Z''_0 & \longrightarrow & B & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow & & & & \downarrow & & \downarrow{\scriptstyle \mathrm{id}} & & \\
0 & \longrightarrow & A & \longrightarrow & Z'_{i-1} & \longrightarrow & \ldots & \longrightarrow & Z'_0 & \longrightarrow & B & \longrightarrow & 0
\end{array}$$

where the middle row is a Yoneda extension as well.

Let $\mathcal{A}$ be an abelian category with objects A, B. Given a Yoneda extension $E : 0 \to A \to Z_{i-1} \to Z_{i-2} \to \ldots \to Z_0 \to B \to 0$ we define an associated element $\delta(E) \in \mathrm{Ext}^i_{\mathcal{A}}(B, A)$ as the morphism $\delta(E) = fs^{-1} : B[0] \to A[i]$ where s is the quasi-isomorphism

$$(\ldots \to 0 \to A \to Z_{i-1} \to \ldots \to Z_0 \to 0 \to \ldots) \longrightarrow B[0]$$

and f is the morphism of complexes

$$(\ldots \to 0 \to A \to Z_{i-1} \to \ldots \to Z_0 \to 0 \to \ldots) \longrightarrow A[i]$$

We call $\delta(E) = fs^{-1}$ the *class* of the Yoneda extension. It turns out that this class characterizes the equivalence class of the Yoneda extension.

Date: stacks-project:derived.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.2. *Let $\mathcal{A}$ be an abelian category with objects A, B and let $i \geq 1$. Any element in $\mathrm{Ext}^i_{\mathcal{A}}(B, A)$ is $\delta(E)$ for some degree i Yoneda extension of B by A. Given two Yoneda extensions E, E' of the same degree then E is equivalent to E' if and only if $\delta(E) = \delta(E')$.*

Proof. Let $\xi : B[0] \to A[i]$ be an element of $\mathrm{Ext}^i_{\mathcal{A}}(B, A)$. We may write $\xi = fs^{-1}$ for some quasi-isomorphism $s : L^\bullet \to B[0]$ and map $f : L^\bullet \to A[i]$. After replacing $L^\bullet$ by $\tau_{\leq 0} L^\bullet$ we may assume that $L^j = 0$ for $j > 0$. Picture

$$\begin{array}{ccccccccc} L^{-i-1} & \longrightarrow & L^{-i} & \longrightarrow & \ldots & \longrightarrow & L^0 & \longrightarrow & B \longrightarrow 0 \\ & & \downarrow & & & & & & \\ & & A & & & & & & \end{array}$$

Then setting $Z_{i-1} = (L^{-i+1} \oplus A)/L^{-i}$ and $Z_j = L^{-j}$ for $j = i-2, \ldots, 0$ we see that we obtain a degree i extension E of B by A whose class $\delta(E)$ equals ξ.

It is immediate from the definitions that equivalent Yoneda extensions have the same class. Suppose that $E : 0 \to A \to Z_{i-1} \to Z_{i-2} \to \ldots \to Z_0 \to B \to 0$ and $E' : 0 \to A \to Z'_{i-1} \to Z'_{i-2} \to \ldots \to Z'_0 \to B \to 0$ are Yoneda extensions with the same class. By construction of $D(\mathcal{A})$ as the localization of $K(\mathcal{A})$ at the set of quasi-isomorphisms, this means there exists a complex $L^\bullet$ and quasi-isomorphisms

$$t : L^\bullet \to (\ldots \to 0 \to A \to Z_{i-1} \to \ldots \to Z_0 \to 0 \to \ldots)$$

and

$$t' : L^\bullet \to (\ldots \to 0 \to A \to Z'_{i-1} \to \ldots \to Z'_0 \to 0 \to \ldots)$$

such that $s \circ t = s' \circ t'$ and $f \circ t = f' \circ t'$, see Categories, Section 04VB. Let E'' be the degree i extension of B by A constructed from the pair $L^\bullet \to B[0]$ and $L^\bullet \to A[i]$ in the first paragraph of the proof. Then the reader sees readily that there exists "morphisms" of degree i Yoneda extensions $E'' \to E$ and $E'' \to E'$ as in the definition of equivalent Yoneda extensions (details omitted). This finishes the proof. $\square$